# NarrativeTrack: A Narrative-Level System for Tracking Misinformation Spread

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## Abstract

Existing misinformation detection tools operate at the level of individual posts or claims, flagging isolated content after it has already spread. This approach fails to capture how false narratives mutate, persist, and evolve through wording changes and reframing over time. This paper presents NarrativeTrack, a software system that tracks misinformation at the narrative level by collecting news articles from multiple sources, generating semantic embeddings using Sentence-BERT, reducing dimensionality with UMAP, and grouping related claims into narrative clusters using DBSCAN. Applied to the COVID-19 lab leak narrative across a two-year period (2021–2022), the system discovered five distinct narrative clusters across 658 news domains and detected the emergence of a previously absent sub-narrative (Legal and Censorship Angle) in 2022—validating the system's ability to detect narrative evolution that post-level tools would miss. An interactive Streamlit dashboard provides visualization of narrative timelines, semantic cluster maps, an article browser, domain analysis, and a narrative mutation view showing vocabulary shifts over time.

## CCS Concepts

• **Information systems** → **Information retrieval**; • **Computing methodologies** → *Natural language processing*.

## Keywords

misinformation tracking, narrative clustering, sentence embeddings, UMAP, DBSCAN, GDELT, information literacy

## 1 Introduction

Information spreads rapidly across digital platforms, and corrective systems such as community notes and crowdsourced fact-checking often respond only after misinformation has already reached large audiences. More critically, these systems evaluate content at the individual post or claim level. When a false narrative is corrected at one source, it frequently resurfaces in a reworded or reframed form that evades detection entirely. The narrative persists not because corrections are absent, but because the unit of analysis is too small.

Consider the COVID-19 lab leak theory. It did not spread as a single uniform claim. It mutated from early fringe speculation about a Wuhan laboratory accident, to accusations about NIH gain-of-function funding, to geopolitical framing by non-Western media, to legal and censorship narratives in 2022. Each mutation represented a distinct sub-narrative, yet all were part of the same evolving ecosystem of misinformation. A system that evaluates posts in isolation cannot detect this structure.

NarrativeTrack addresses this gap by treating misinformation as an evolving narrative ecosystem. The system collects textual data at scale, groups semantically related claims into narrative clusters using NLP techniques, and visualizes how those clusters grow, decay, and give rise to new sub-narratives over time. This provides a qualitatively different kind of situational awareness from existing tools: not just whether a claim is false, but how, when, and through which sources it mutates and amplifies.

The intended users are researchers, journalists, and information-literate general users who want visibility into how misinformation develops, not just whether individual claims have been labeled.

## 2 Related Work

### 2.1 Post-Level and Claim-Level Systems

The dominant paradigm in deployed misinformation detection operates at the level of individual posts or claims. Community Notes (formerly Birdwatch) on X (formerly Twitter) allows users to annotate individual posts with contextual notes. PolitiFact and Snopes evaluate individual claims submitted by editors or users. ClaimBuster [1] provides automated check-worthiness detection at the sentence level. These systems share a common limitation: they evaluate content atomically, with no model of how claims relate to one another across sources or time.

These tools also tend to be reactive. Research has shown that corrections rarely reach the same audience as the original misinformation [2], a problem compounded when the narrative has already mutated into new forms.

### 2.2 Narrative and Cluster-Level Approaches

Vosoughi et al. [3] studied the spread of true and false news on Twitter as cascades, finding that false news diffuses faster and more broadly. However, this work tracked individual stories rather than the evolution of narrative variants. SemEval shared tasks on rumour detection [4] have explored claim stance and veracity, but without modeling how claims mutate over time. NarrativeTrack differs in its explicit focus on semantic similarity between claims, temporal windowing to capture evolution, and interactive visualization of narrative lifecycle.

### 2.3 Data Sources

Prior work has relied heavily on Twitter data. However, Twitter's API became paywalled in 2023. Reddit was an alternative, but self-service API access was discontinued in late 2025. GDELT (Global Database of Events, Language and Tone) [5] offers a free, large-scale alternative covering thousands of news outlets globally. To our knowledge, NarrativeTrack is among the first systems to use GDELT specifically for narrative-level misinformation tracking.

## 3 System Overview

NarrativeTrack consists of five pipeline stages and an interactive dashboard. Table 1 summarizes the architecture.

**Table 1: NarrativeTrack pipeline stages and outputs.**

| Stage | Component | Output |
|---|---|---|
| 1 | Data Collection | Raw article JSON/CSV by month |
| 2 | Text Cleaning | Filtered, normalized, deduplicated text |
| 3 | Embedding | 384-dimensional semantic vectors |
| 4 | Clustering | Narrative cluster assignments |
| 5 | Dashboard | Interactive Streamlit interface |

All pipeline stages run in a single Google Colab notebook, with data persisted to Google Drive between sessions. The system requires no API keys and no local storage beyond the Colab runtime.

## 4 Implementation

### 4.1 Data Source Selection and Pivots

The data source for this system was not fixed from the outset. The original proposal assumed collection from Twitter/X. This plan was abandoned when Twitter's API became paywalled in 2023, with developer access requiring a minimum of $100 per month.

The project then pivoted to Reddit, which offered free API access via the PRAW library. However, in November 2025, Reddit implemented its Responsible Builder Policy, ending self-service API credential provisioning. New API access requires manual review with a stated seven-day response target—not feasible for the project timeline.

The final pivot was to **GDELT (Global Database of Events, Language and Tone)**, which monitors over 100,000 news outlets across 100 languages and exposes a free query API requiring no authentication. Crucially for narrative-level tracking, GDELT covers mainstream journalism rather than social media, making it well suited for studying how narratives migrate from fringe to mainstream coverage.

### 4.2 Data Collection

The collection module queries GDELT's Doc 2.0 API in monthly windows. Monthly windowing is a deliberate design decision: it enables temporal analysis of narrative evolution by preserving the time bucket in which each article was collected, and each window fetches up to 250 articles (the GDELT API maximum per request).

For the case study, the query *"covid lab leak"* was used with English-language filtering (`sourcelang:eng`), collecting across January 2021 to January 2023. GDELT rate limiting (HTTP 429) affected approximately half the monthly windows, resulting in 8 of 25 target windows returning data and a total of 1,784 articles collected.

### 4.3 Text Cleaning and Filtering

GDELT's broad keyword matching returned substantial noise. Three cleaning steps were applied:

(1) **Relevance filtering**: articles were required to contain at least one of 16 hand-curated keywords directly associated with the lab leak narrative (e.g., *lab leak*, *gain of function*, *wuhan institute*). This reduced the dataset from 1,784 to 384 articles.
(2) **Text normalization**: titles were lowercased, URLs removed, and special characters stripped.
(3) **Deduplication**: articles with identical normalized titles were removed, producing 239 unique articles across 658 domains.

### 4.4 Semantic Embedding

Each article title was encoded as a 384-dimensional dense vector using the `all-MiniLM-L6-v2` Sentence-BERT model [6]. This model was selected for its optimization on semantic similarity tasks for short texts, its lightweight footprint ( 90 MB), and its ability to group semantically equivalent claims that share few exact keywords.

### 4.5 Dimensionality Reduction and Clustering

Clustering proceeded in two steps. First, UMAP [7] reduced the 384-dimensional embeddings to 10 dimensions while preserving local semantic structure. This step was critical: initial DBSCAN on raw normalized embeddings (eps=0.5) clustered only 43 of 239 articles (82% noise). UMAP reduction to 10 dimensions resolved this, producing coherent clusters with 0% noise at eps=0.7.

DBSCAN [8] was chosen over k-means because it does not require specifying the number of clusters in advance, and naturally handles outliers. With eps=0.7 and min_samples=2 on the 10-dimensional UMAP output, DBSCAN produced five coherent narrative clusters. A separate 2D UMAP projection was computed for visualization only.

### 4.6 Interactive Dashboard

The dashboard was implemented in Streamlit and exposed via ngrok tunneling from Google Colab. It provides five tabbed views:

- **Timeline**: stacked area and line charts showing article volume per cluster by month.
- **Cluster Map**: 2D UMAP scatter plot with hover tooltips.
- **Article Browser**: filterable table with links to original sources.
- **Domain Analysis**: top contributing domains overall and per cluster.
- **Narrative Mutation**: TF-IDF heatmap of distinctive terms per month within a selected cluster, plus an Early vs. Late vocabulary shift panel.

## 5 Evaluation

### 5.1 Evaluation Approach

The system was evaluated through a case study on the COVID-19 lab leak narrative. Evaluation criteria were: (1) whether clusters group semantically related articles, (2) whether the system detects meaningful narrative changes over time, and (3) whether the visualization is interpretable by a non-technical user.

### 5.2 Cluster Quality

The five clusters correspond to identifiable real-world narrative variants, shown in Table 2.

Manual inspection confirms that the groupings reflect real narrative relationships. Clusters 1 and 2, while both related to the lab

Table 2: Narrative clusters discovered by NarrativeTrack.

| Narrative | Articles | First Seen | Peak |
|---|---|---|---|
| Wuhan Lab / China Cover-up | 113 | 2021-01 | 2021-01 |
| Fauci & NIH Funding | 72 | 2021-01 | 2021-09 |
| Congressional Investigation | 36 | 2021-01 | 2021-01 |
| Geopolitics Framing | 7 | 2021-09 | 2021-09 |
| Legal & Censorship Angle | 11 | 2022-04 | 2022-09 |

leak narrative, are semantically and factually distinct sub-narratives that a keyword-based system would group together.

## 5.3 Narrative Evolution Detection

The most significant finding is the emergence of the Legal and Censorship Angle cluster, which first appears in April 2022 and contains no articles from 2021. This cluster includes articles about state attorneys general filing suits against Biden administration officials and Fauci for alleged social media collusion. This represents genuine narrative evolution: by 2022, the lab leak story had partially transformed from a scientific debate into a free speech and legal argument. NarrativeTrack detected this transition automatically, validating the system's central claim.

## 5.4 Limitations

The dataset covers only 8 of 25 target months due to rate limiting. Clustering uses article titles rather than full text, as GDELT does not return body content. The evaluation is qualitative; a quantitative comparison against labeled ground truth would strengthen conclusions.

# 6 Challenges and Design Decisions

## 6.1 Data Source Availability

The two most commonly used social media data sources in misinformation research—Twitter and Reddit—became unavailable during the course of this project due to API policy changes. This required two full implementation pivots and significant unplanned investigation time. The final choice of GDELT, while not the original plan, proved to be a stronger fit for narrative-level analysis due to its coverage of professional journalism and absence of authentication requirements.

## 6.2 Clustering Parameter Sensitivity

DBSCAN's performance is sensitive to the eps parameter. The initial configuration (eps=0.5 on raw normalized embeddings) produced 82% noise. Introducing UMAP as a preprocessing step fundamentally changed the geometry of the embedding space. The final configuration—10-dimensional UMAP followed by DBSCAN at eps=0.7—required iterative experimentation and highlights that narrative clustering on semantic embeddings is a non-trivial tuning problem.

## 6.3 High-Dimensional Visualization

PCA-based scatter plots showed no visible cluster structure because PCA maximizes variance rather than preserving local neighborhood relationships. Switching to UMAP for the 2D visualization resolved this. The distinction between the 10D UMAP used for clustering and the separate 2D UMAP used for visualization is an important implementation detail that significantly affected output quality.

# 7 Software Documentation

## 7.1 Requirements

NarrativeTrack runs entirely in Google Colab. No local installation is required. A Google account with Google Drive access is sufficient. An ngrok account (free tier) is required to expose the dashboard.

## 7.2 Running the Pipeline

(1) Open `pipeline.ipynb` in Google Colab.
(2) Edit the Config cell: set `QUERY`, `START_DATE`, `END_DATE`.
(3) Run Step 1 (Mount Drive) and grant Google Drive access.
(4) Run Step 2 (Install Dependencies). Takes ≈60 seconds.
(5) Run Step 3 (Collect). Output: article counts per monthly window, JSON and CSV saved to Drive.
(6) Run Step 4 (Clean). Output: before/after article counts.
(7) Run Step 5 (Embed). Output: confirmation of `(N, 384)` embedding shape.
(8) Run Step 6 (Cluster). Output: cluster count, noise %, and representative titles per cluster.
(9) Run Step 7 (Timeline). Output: matplotlib charts saved to Drive.
(10) Run Steps 8–10 (Dashboard). Output: a public ngrok URL to open in any browser.

## 7.3 Configuration

Key parameters and their defaults are shown in Table 3.

Table 3: Configurable parameters and defaults.

| Parameter | Default | Description |
|---|---|---|
| `QUERY` | `"covid lab leak"` | GDELT search query |
| `START_DATE` | `"20210101"` | Start date (YYYYMMDD) |
| `END_DATE` | `"20230101"` | End date (YYYYMMDD) |
| `eps` | `0.7` | DBSCAN cluster radius |

## 7.4 Source Code

Source code is available at: https://github.com/if3ol/CS510/blob/main/pipeline_(2).ipynb. The repository includes `pipeline.ipynb` and `requirements.txt`. Data files are not included and must be generated by running the notebook.

# 8 Conclusion

NarrativeTrack demonstrates that narrative-level misinformation tracking is both technically feasible and empirically informative on real-world data. Applied to the COVID-19 lab leak narrative, the system identified five semantically coherent narrative clusters and automatically detected the emergence of a new sub-narrative in 2022. This finding directly supports the central motivation: post-level tools, by evaluating content in isolation, miss the structural evolution of misinformation over time.

Future work includes implementing exponential backoff in the collector to fill missing temporal windows, extending the pipeline to full article text, applying the system to additional misinformation narratives, and developing a quantitative evaluation framework using labeled claim datasets.

## 9 Team Contribution

This project was completed individually. Ifeoluwa Olusola (olusola4) designed and implemented all components of the system, including the data collection pipeline, text cleaning, semantic embedding, UMAP dimensionality reduction, DBSCAN clustering, timeline visualization, and the Streamlit dashboard with all five tabs. All writing, evaluation, and documentation were completed by the same author.

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